

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered a pluggable distributed executor routing pipeline nodes across Docker and AWS Lambda with an S3 artifact store, eliminating redundant recomputation under scale constraints.
- Provisioned and operated multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA hardware and InfiniBand/RoCE networking for high-throughput, low-latency serving.
- Own production services end-to-end, architecture through deployment, monitoring, and on-call, handling millions of daily transactions at 99%+ uptime.
- Replaced polling with SSE for live metrics streaming, cutting redundant API calls 90% through performance-focused architectural changes.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Extended and optimized inference serving internals with kernel-level development (CUDA/Triton), distributed KV cache, and speculative decoding to cut compute overhead.
- Cut p99 serving latency 420ms to 140ms across 50K+ daily requests through low-level performance analysis and an async serving redesign with autoscaling.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production C++ and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning at scale.
- Standardized failure-handling contracts across 25+ services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus distributed storage engine from scratch in C++: concurrent writes, linearizable reads, and log compaction, a hands-on study of storage-internals and consensus tradeoffs under load.
- Built a companion VPC control plane simulator modeling subnet routing and resource placement over the same distributed substrate.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Full-stack platform with FastAPI backend, React/TypeScript dashboard, and PostgreSQL, exposing APIs for rubric-based scoring and latency tracking at scale.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: C++, Java, Python, Go, Rust, SQL

Systems & Storage: Distributed Systems, Consensus Protocols, Storage Internals, Fault-Tolerant Scheduling, Concurrency

Performance: CUDA/Triton Kernel Development, Low-Level Performance Analysis, Distributed KV Cache, Latency Optimization

Infrastructure: Multi-Cloud GPU Clusters (AWS, GCP, CoreWeave), InfiniBand/RoCE, Docker, Kubernetes, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer